8.17 $S_0 = 120$

P_0, C_0 $K = 110$, $T = 1$

10 5

$r = 5\%$

- forwd / prepaid fwd

Put-Call parity

$$P_0 + PV(F_{0T}) = C_0 + PV(K)$$

$$P_0 + F_{0T}^{prep} = C_0 + PV(K)$$

- $$10 + F_{0T}^{prep} = 5 + \frac{110}{e^{0.05}}$$

$$F_{0T}^{prepaid} = 5 - 10 + \frac{110}{e^{0.05}} = \$99.64$$

- forward. $F_{0T} = e^{rT} F_{0T}^{prepaid}$

$$= e^{0.05} \cdot 99.64 \simeq \$104.74$$

- dividend $F_{0T}^{prepaid} = S_0 \cdot e^{-\delta T}$

$$e^{-\delta \cdot 1} = \frac{F_{0T}^{prepaid}}{S_0} \rightarrow \ln(e^{-\delta}) = -\delta$$

$$= \ln\left(\frac{99.64}{120}\right)$$

$$\delta = 18\%$$

FORWARD Trades @ 110

↑

Short mkt fwd

long P-C forward

$$\underbrace{C}_{\text{long } C} \underbrace{- P}_{\text{short } P} \underbrace{+ PV(K)}_{\text{long bond}} = \underbrace{PV(F_{0T})}_{\text{long}}$$

		$t = 0$	$t = T = 1$	
			$S_T < K$	$S_T > K$
3	long call	$-C_0 = -5$	0 (OTM)	$S_T - K = S_T - 110$
	long bond	$-PV(K) = -\frac{110}{e^{0.05 \cdot 1}} =$	$+110 = K$	$+110$
	short put	$+P_0 = +\$10$	$-110 + S_T$	0
		$-5 + 10 - \frac{110}{e^{0.05}} = -99.635$	$+S_T$	$+S_T$
1	short mkt fwd	0	$-S_T$	$+110$
	borrow	$+\frac{110}{e^{0.05}} \simeq 104.635$	-110	
		$+\$5$	0	

Arbitrage.